

BENGAL COLLEGE OF ENGINEERING AND TECHNOLOGY

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TOPIC

MAXIMIZE OBJECTIVE FUNCTION PROBLEM

Subtitle:

A Mathematical Optimization Technique

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INTRODUCTION

Definition:- Maximizing an objective function is the process of finding the best value (maximum) of an objective function, subject to given constraints.

Applications: Operations research, economics, business decision-making, production planning, etc.

Problem Statement

Given an objective function $f(x)$ and a set of constraints, find the values of x that maximize $f(x)$.

Equation: Maximize $f(x) = \dots\dots\dots$

Types of Optimization Problems

- Linear Programming (LP)
- Nonlinear Programming (NLP)
- Integer Programming (IP)
- Mixed-Integer Programming (MIP)

Objective Function

- **Definition:** A mathematical expression that needs to be maximized or minimized.
It typically represents a goal like profit, cost, or resource allocation.

Example:

$$Z = 5x_1 + 4x_2$$

where Z is the objective function to maximize, and
 x and y are decision variables.

Constraints

- Constraints are limitations on the values of the decision variables
- Equality constraints ($h(x) = 0$)
- Inequality constraints ($g(x) \leq 0$)

Example:

$$x_1 + x_2 \leq 10 \text{ (Resource constraint).}$$

$$x_1, x_2 \geq 0 \text{ (Non-negativity constraint)}$$

Optimization Techniques

1. Linear Programming (LP) methods
2. Nonlinear Programming (NLP) methods
3. Integer Programming (IP) methods

1. Linear Programming (LP) Methods

- LP methods are used to solve linear optimization problems
- Simplex method
- Dual simplex method
- Interior-point method

Linear Programming (LP): A mathematical method used to maximize (or minimize) a linear objective function subject to linear constraints.

General Form: **Maximize:** $Z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$

Subject to: $A_1 x_1 + A_2 x_2 + \dots + A_n x_n \leq b$

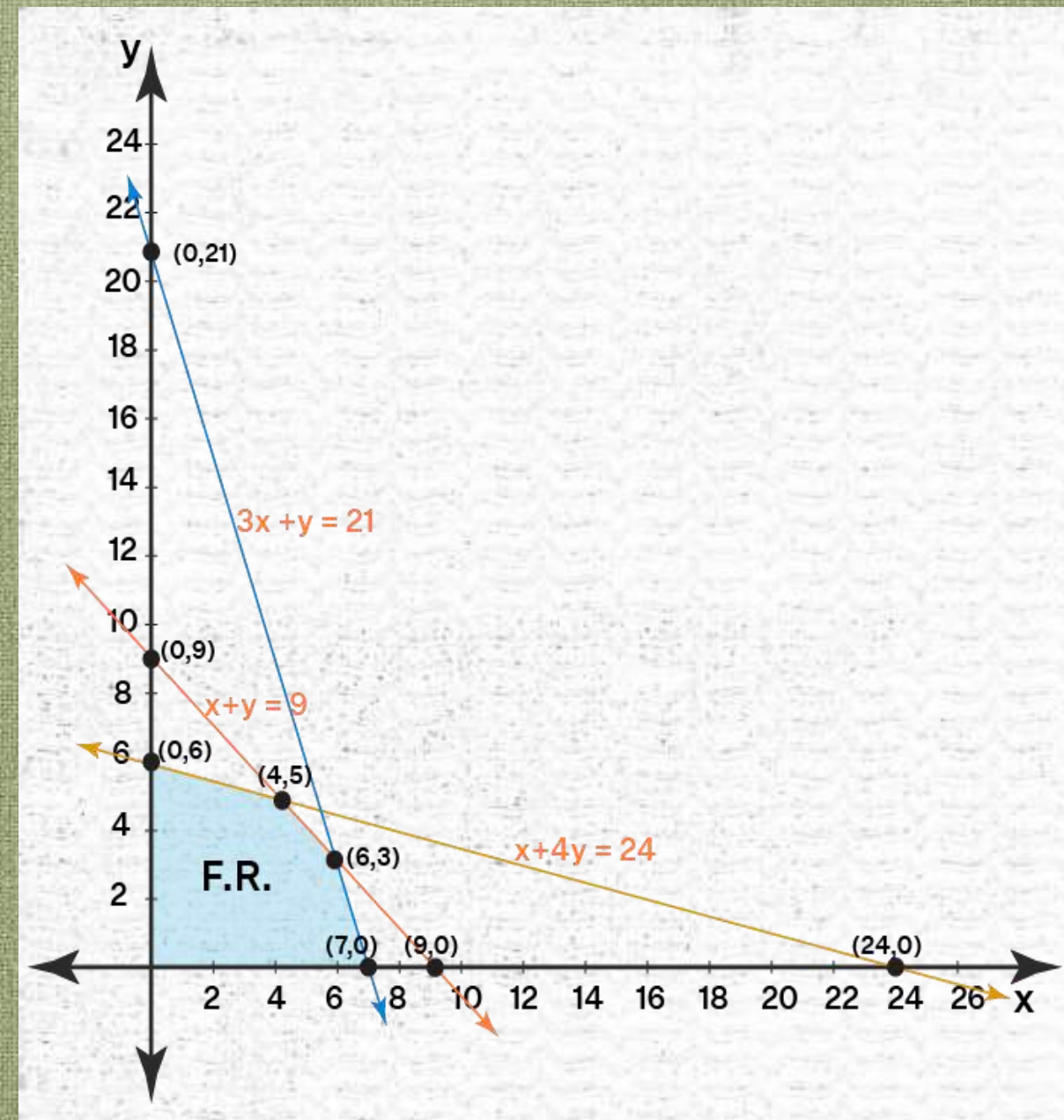
$x_1, x_2, \dots, x_n \geq 0$

2. Nonlinear Programming (NLP) Method

- Nonlinear Programming (NLP) Methods
- NLP methods are used to solve nonlinear optimization problems
- Gradient-based methods
- Quasi-Newton methods
- Conjugate gradient methods

3. Integer Programming (IP) Methods

- IP methods are used to solve integer optimization problems
- Bullet points:
- Branch-and-bound method
- Cutting-plane method



Graphical Method (2 Variables)

- **Method:** Use when there are only two decision variables. The feasible region is plotted, and the objective function is maximized at a corner point.
- **Steps:**
 1. Plot the constraints on a graph.
 2. Identify the feasible region.
 3. Evaluate the objective function at the corner points of the feasible region.

Types of Graphical Method

1. Bounded Feasible Solution
2. Unbounded Feasible Solution
3. No Feasible Solution

Simplex Method (For More Variables)

- **Method:** The Simplex Method is an iterative algorithm used to solve linear programming problems with more than two variables.

Steps:

1. Convert the problem to standard form (with equality constraints).
2. Set up the initial tableau.
3. Perform iterations to improve the solution until reaching optimality. **Technical**

Terminology:-

- **Basic variable:** Those variable which coefficient are 1 in one equation and 0 in other equation.
- **Non-Basic variable:** Those are the variable whose coefficient are taken any value whether it is +ve / -ve
- **Slack variable:**
 - a. If Inequality is less than ($<$) then, we have to ADD variable to Balance. {Slack variable}.
 - b. If Inequality is greater than ($>$) then, we have to SUB. Variable to Balance {Surplus Variable}.
 - c. If equation is equal ($=$) then, we have to do nothing with variable.

1. Example Problem: (BY GRAPHICAL & SIMPLEX METHOD)

Objective Function: MAX (Maximize $Z=3x_1+2x_2$)

Subject to Constraints: $x_1 + x_2 \leq 5.$

$$x_1 - x_2 \geq 1$$

$$x_1, x_2 \geq 0$$

Using the Graphical Method

1. Solution Using the Graphical Method

Step-by-Step Approach:

Step 1: Plot the Constraints

We need to plot each constraint on a graph.

- **For.** $x_1 + x_2 \leq 5$

Rewrite it as $x_2 = 5 - x_1$ to get the equation of the line.

- **For.** $x_1 - x_2 \geq 1$:
Plot this line and shade the region that satisfies $x_1 + x_2 \leq 5$.

Rewrite it as $x_2 = x_1 - 1$ to get the equation of the line.

Plot this line and shade the region that satisfies $x_1 - x_2 \geq 1$.

- **For.** $x_1 \geq 0$ and $x_2 \geq 0$:

These constraints mean that the solution will be in the first quadrant of the graph (where both x and y are non-negative).

Step 2: Identify the Feasible Region

The feasible region is the area where all the shaded regions from the constraints overlap. It will be a polygon, and the optimal solution lies at one of the corner points of this region.

Step 3: Evaluate the Objective Function

Evaluate $Z = 3x_1 + 2x_2$ at each corner point of the feasible region. The corner points can be obtained by finding the intersections of the lines:

- Intersection of $x_1 + x_2 = 5$ and $x_1 - x_2 = 1$
- Intersection of $x_1 + x_2 = 5$ and $x_1 = 0$
- Intersection of $x_1 - x_2 = 1$ and $x_2 = 0$

Step 4: Calculate the Maximum Value

Evaluate Z at each corner and find which one gives the highest value of Z.

Using the Simplex Method

2. Solution Using the Simplex Method

For this example, we will convert the problem to standard form and solve it using the **Simplex Method**.

Step 1: Convert the Problem to Standard Form

The constraints are:

1. $x_1 + x_2 \leq 5$
2. $x_1 - x_2 \geq 1$
3. $x_1 \geq 0$
4. $x_2 \geq 0$

For the Simplex method, we need to convert the inequalities to equalities by introducing **slack** and **surplus** variables:

1. $x_1 + x_2 + s_1 = 5$ (Introduce slack variable s_1 for the first constraint)
2. $x_1 - x_2 - s_2 = 1$ (Introduce surplus variable s_2 for the second constraint)
3. $x_1 \geq 0$
4. $x_2 \geq 0$

Now, we need to maximize the objective function:

$$Z = 3x_1 + 2x_2$$

We introduce artificial variables s_1 and s_2 into the objective function for the standard form.

Step 2: Set Up the Initial Simplex Tableau

The initial tableau will look like this:

C _B i	Basic variables	X ₁	X ₂	S ₁	S ₂	SOLUTION	RATIO
0	s ₁	1	1	1	0	5	
0	s ₂	1	-1	0	-1	1	
	z	-3	-2	0	0	0	

Here X_1 and X_2 are decision variables & S_1 , S_2 are slack variable.

The Condition Optimality For Simplex Method $[C_j - Z_j \leq 0]$.

Step 3:-
$$Z_j = \sum_{i=1}^2 (C_{B_i})x(a_{ij})$$

Step 4: Find Max. Key value

Step 5: Now Find Key Column

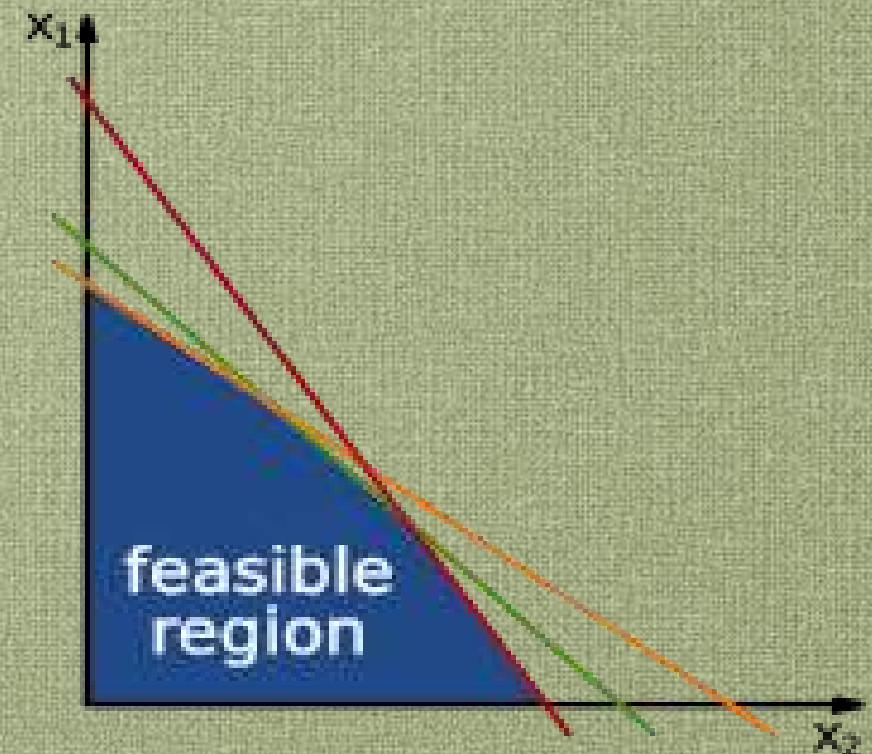
Step 6: Find Ratio [solution/key column]

Step 7:- Select pivot Element:- In simplex in a OT a particular value when is interesting point of maximization and minimization for specific value is called pivot value for a specific simplex method

Step 8: Perform Simplex Iterations

The **Simplex Method** will now proceed through a series of steps where we look for the most negative value in the last row (excluding the RHS column) and perform pivot operations to get closer to the optimal solution.

Each iteration moves towards an optimal solution, and you will continue until there are no more negative values in the objective row.



Applications in Real Life

- **Business:** Maximizing profit, minimizing costs in production.
- **Economics:** Optimizing resource allocation in markets.
- **Logistics:** Route optimization, supply chain management.
- **Healthcare:** Maximizing efficiency in medical treatment planning.
- **Define the Objective Function**
- **Identify Constraints**
- **Choose an Optimization Method**

Conclusion

- **Summary:** Maximizing an objective function involves finding the optimal solution while adhering to constraints.
 - **Methods:** Graphical method (for two variables), Simplex method (for more variables).
 - **Importance:** Plays a crucial role in optimization problems across various industries and sectors.
-
- The Maximize Objective Function Problem is a mathematical optimization Technique used to find the optimal solution to a problem
 - Summary of key points
 - Image: a graph or chart showing an optimization problem

REFERENCES

- *Introduction to Operations Research* by Hillier & Lieberman
- *Operations Research: An Introduction* by Taha H.A.
- *Numerical Optimization* by Norendal & Wright
- *Optimization by Vector Space Methods* by Lemberger

Online resource: [Coursera: Nonlinear Programming](#)

Online resource: Google , WIKIPEDIA etc.



THANK YOU

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